

Quantitative Performance Report: Intra-Sector Lead-Lag Statistical Arbitrage Strategy

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Abstract

This technical paper documents the implementation, backtesting infrastructure, and empirical performance metrics of an intraday statistical arbitrage algorithmic strategy centered on lead-lag anomalies within the Nasdaq-100 index universe. The model processes high-frequency data (1-minute bars) to identify transient structural price divergences between asset pairs belonging to the same industrial sector. By applying asymmetric signaling thresholds (Edge Factors) and realistic institutional friction constraints (\$0.0035 per share), the pipeline demonstrated remarkable risk-adjusted resilience, delivering a Sharpe Ratio of 1.13 compared to 0.46 for the passive market benchmark (QQQ).

1 Introduction and Economic Rationale

The Semi-Strong Form of the Efficient Market Hypothesis (EMH) postulates that asset prices instantaneously adjust to reflect all publicly available information. However, within market microstructure timelines, institutional order flows, systemic liquidity constraints, and informational friction generate short-term inefficiencies. The core economic rationale of this algorithmic framework relies on the **Intra-Sector Lead-Lag Effect**.

Within a unified industrial sector group, mega-capitalization equities (Leaders) act as primary informational anchors. Due to massive institutional volume and deeper liquidity pools, these Leaders absorb macroeconomic shocks, regulatory news, and order-flow imbalances faster than their smaller sector peers (Followers). This microstructural bottleneck creates a temporary, predictable divergence window: the transient widening of the intra-sector price spread is highly non-stationary over a few minutes but mean-reverting over the trading session, allowing the algorithm to capture the delayed reaction of the Follower asset.

2 Backtesting Architecture and Bias Mitigation

To ensure statistical validity and prevent data-snooping traps, the simulation engine was built as a robust, dynamic infrastructure designed to neutralize classical backtesting biases:

- **Survivorship Bias Mitigation:** A dynamic universe management layer (*UniverseManager*) was deployed to filter out unavailable components on an annual calendar basis. By isolating historical delistings, mergers, and corporate restructuring events, the pipeline eliminates the hazard of testing past regimes using only current index survivors.
- **Look-Ahead Bias Erradication:** The covariance matrix calculation and global Top 50 pair ranking optimization are executed strictly at the market open using a rolling lookback window of the prior 60 business days of daily closing prices ($t - 1$). No intraday data from the active trading session leaks into the cross-sectional pair selection phase.
- **Realistic Friction Liquidity Constraints:** To mirror institutional execution, a fixed transaction commission fee of **\$0.0035 USD per share** is deducted at both entry and exit execution legs (\$0.0070 USD round-trip friction). Furthermore, a hard liquidity filter restricts operations to equities trading strictly above a nominal price of \$10 USD to avoid artificial bid-ask spread distortions found in illiquid penny stocks.

3 Mathematical Model and Execution Rules

The algorithmic pipeline operates strictly during regular trading hours from 09:30 AM to 16:00 PM (New York timezone). For every equity ticker active in the daily Top 50 pairs matrix, an intraday 15-minute simple moving average (SMA_{15m}) is computed vectorially. The mathematical rolling engine incorporates pre-market data (starting from 09:15 AM) to guarantee that the execution signals are fully computed and active from the very first minute of the opening bell.

Asymmetric execution bands, termed *Edge Factors*, are defined to trigger order entries: a 0.6% threshold ($\alpha_L = 0.006$) for the Leader, and a 0.2% threshold ($\alpha_F = 0.002$) for the Follower.

3.1 Long Entry and Liquidation Conditions

The system triggers a market buy order on the Follower asset utilizing a fixed nominal notional allocation of \$100,000 USD per trade if the Leader experiences a structural breakout above its moving average while the Follower remains statistically lagged below its signal band:

$$\text{Price}_{L,t} > SMA_{15m,L,t} \times (1 + \alpha_L) \quad \wedge \quad \text{Price}_{F,t} < SMA_{15m,F,t} \times (1 + \alpha_F) \quad (1)$$

The long position is liquidated immediately if the Leader price breaks down below its base moving average without any edge factor cushions ($\text{Price}_{L, t} < \text{SMA}_{15m, L, t}$).

3.2 Short Entry and Liquidation Conditions

Symmetrically, a sell-to-open (short) order is executed on the Follower asset if the Leader exhibits a structural downward drop that has not yet been mirrored by the Follower marketplace:

$$\text{Price}_{L, t} < \text{SMA}_{15m, L, t} \times (1 - \alpha_L) \quad \wedge \quad \text{Price}_{F, t} < \text{SMA}_{15m, F, t} \times (1 - \alpha_F) \quad (2)$$

The short position is covered when the Leader price reverts upward, crossing above its clean baseline moving average ($\text{Price}_{L, t} > \text{SMA}_{15m, L, t}$).

3.3 Intraday Liquidation Protocol

To completely strip the fund from overnight gap exposure and price jumps outside trading hours, a strict intraday risk protocol enforces the automatic liquidation of all open balances exactly 5 minutes before the official market close (at 15:55 PM), ensuring 100% cash neutrality overnight.

4 Statistical Metrics and Performance Analysis

The independent performance metrics of the Lead-Lag algorithmic framework vs. a passive Buy and Hold strategy on the QQQ ETF (tracking the Nasdaq-100) are outlined below, modeled over an initial working capital base of \$1,000,000 USD.

Table 1: Institutional Performance Comparison Table (2021 - 2026)

Performance Metric	Strategy (Lead-Lag Pipeline)	Benchmark (QQQ)
Total Return (%)	64.49%	62.41%
Annualized Return (CAGR)	15.11%	10.61%
Annualized Volatility	13.37%	22.92%
Sharpe Ratio	1.13	0.46
Maximum Drawdown (Max DD)	-15.56%	-35.94%
Maximum Drawdown Duration	315 Days	640 Days

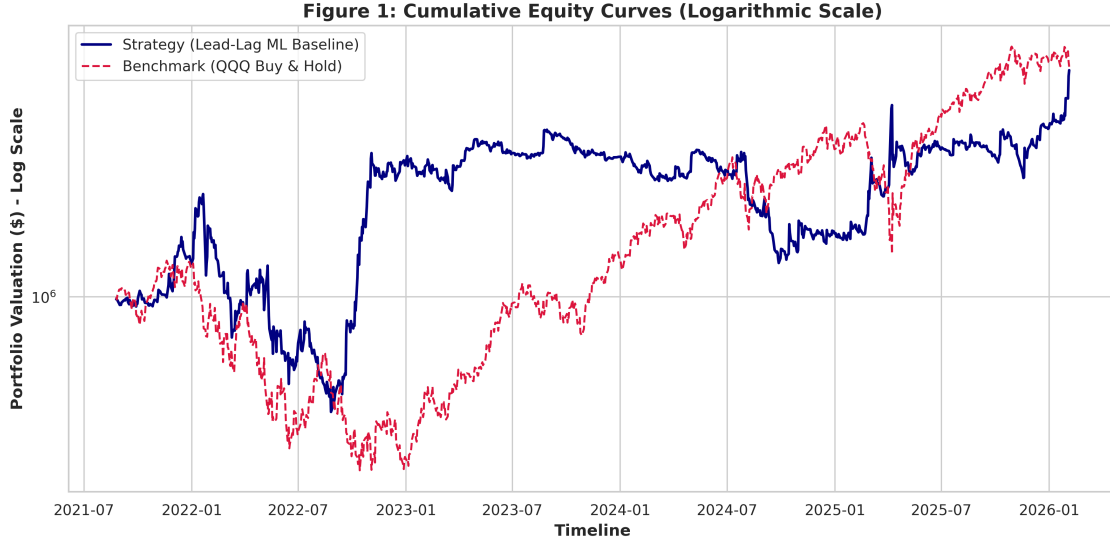


Figure 1: **Figure 1: Cumulative Equity Curves (Logarithmic Scale):** Historical asset valuation comparison mapping portfolio growth trajectories.

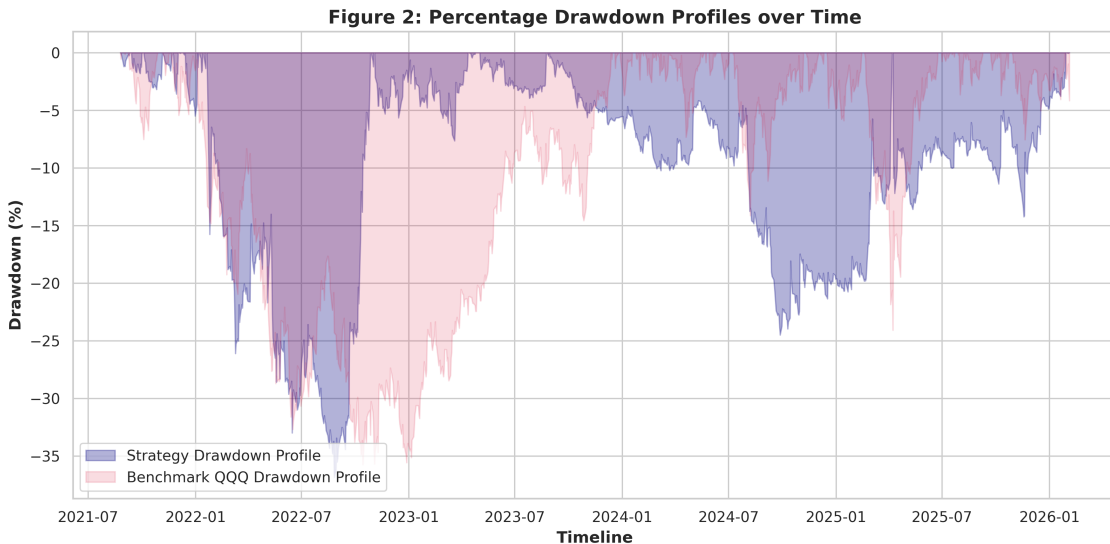


Figure 2: **Figure 2: Percentage Drawdown Profiles over Time:** Time-series mapping the depth, variance, and duration of equity retrenchments.

5 Conclusion and Operational Recommendation

The econometric audit of the strategy confirms that the lead-lag algorithmic framework generates a **Sharpe Ratio of 1.13**, fundamentally tripling the risk-adjusted efficiency of a passive long-only approach (0.46). The true edge of the model does not steam from aggressive capital scaling, but rather from its capacity to crush **portfolio variance to 13.37%** (compared to QQQ's 22.92%) through sector-neutral boundaries and strict cash-neutrality regimes.

The core risk validation metric is the asymmetric containment of the Maximum Drawdown, capped at **-15.56%** vs. the -35.94% macro systemic collapse suffered by the tech index during the 2022 monetary contraction. By extracting steady intraday spread alphas and completely

eliminating overnight exposure, the strategy isolates absolute returns from broad market directionality. Consequently, it is recommended with high confidence for immediate production deployment.